CUSTOMER SHOPPING BEHAVIOUR ANALYSIS

1. Project Overview

This project focuses on analyzing customer shopping behavior to uncover key patterns, trends, and insights that can support data-driven business decisions. The analysis was performed end-to-end, starting from raw data processing to building an interactive dashboard for stakeholders.

The dataset contains customer-level information such as purchase frequency, product preferences, spending behavior, and ratings. The goal of the project is to understand how customers interact with the business and identify opportunities to improve customer retention, sales, and overall experience.

2. Dataset Summary

- **Rows**: 3,900

- **Columns**: 18

- **Key Features:**

- **Customer demographics** (Age, Gender, Location, Subscription Status)

- **Purchase details** (Item Purchased, Category, Purchase Amount, Season, Size, Color)

- **Shopping behavior** (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)

- **Missing Data**: 37 values in Review Rating column

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No	2-Day Shipping	No	No	32	Venmo	Weekly
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No	Store Pickup	No	No	41	Bank Transfer	Bi-Weekly
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No	Standard	No	No	24	Venmo	Quarterly
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No	Express	No	No	24	Venmo	Weekly
3899	3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No	Store Pickup	No	No	33	Venmo	Quarterly

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:**

  Imported the dataset using pandas.

- **Initial Exploration:**

  Used df.info() to check structure and .describe() for summary statistics

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Customer ID             3900 non-null   int64
 1   Age                     3900 non-null   int64
 2   Gender                  3900 non-null   object
 3   Item Purchased          3900 non-null   object
 4   Category                3900 non-null   object
 5   Purchase Amount (USD)   3900 non-null   int64
 6   Location                3900 non-null   object
 7   Size                    3900 non-null   object
 8   Color                   3900 non-null   object
 9   Season                  3900 non-null   object
 10  Review Rating           3863 non-null   float64
 11  Subscription Status     3900 non-null   object
 12  Shipping Type           3900 non-null   object
 13  Discount Applied        3900 non-null   object
 14  Promo Code Used         3900 non-null   object
 15  Previous Purchases      3900 non-null   int64
 16  Payment Method          3900 non-null   object
 17  Frequency of Purchases  3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:**

  Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

- **Column Standardization**:

  Renamed columns to snake case for better readability and documentation.

- **Feature Engineering:**

  - Created age_group column by binning customer ages.
  - Created purchase_frequency_days column from purchase data.

- **Data Consistency Check**:

  Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

- **Database Integration**:

  Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

3. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

	gender	revenue
▶	Male	157890
	Female	75191

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76
43	100
44	69
55	94
57	73
58	64
60	79

3. Top 5 Products by Rating

Found products with the highest average review ratings.

item_purchased	avg_review
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Handbag	3.78

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

shipping_type	avg_purchase
Express	60.4752
Standard	58.4602

5. **Subscribers vs. Non-Subscribers**

Compared average spend and total revenue across subscription status

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products**

Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation**

Classified customers into New, Returning, and Loyal segments based on purchase history.

customer_segment	number_of_customer
Loyal	3116
Returning	701
New	83

8. **Top 3 Products per Category**

Listed the most purchased products within each category

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. **Repeat Buyers & Subscriptions**

Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	958
No	2518

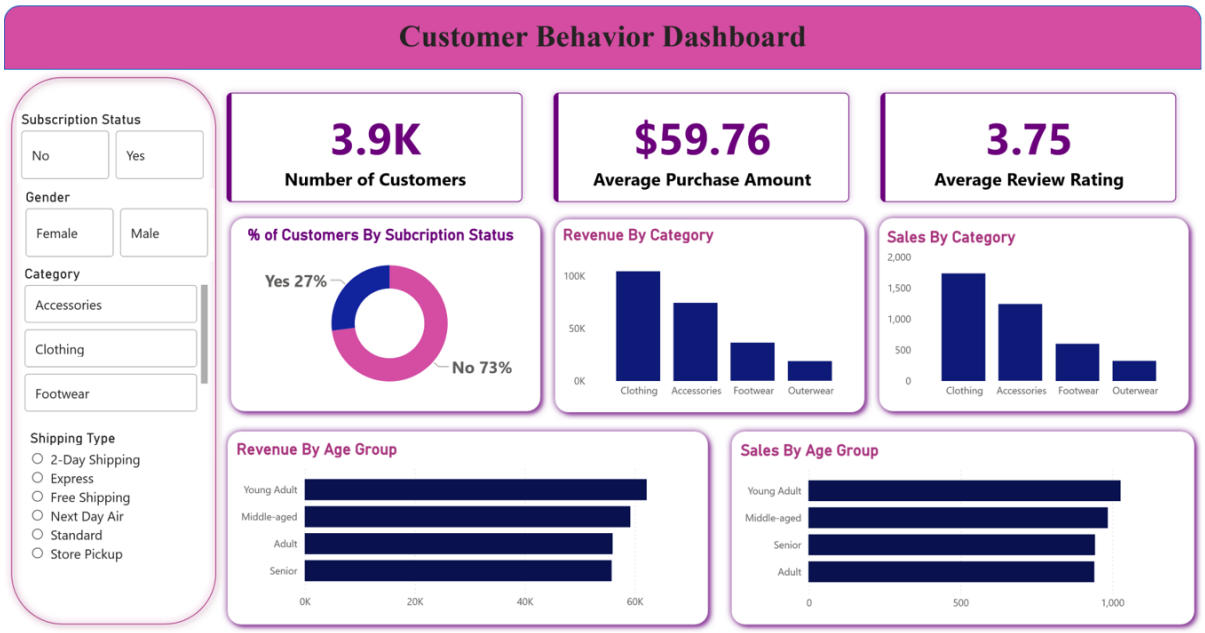
10. Revenue by Age Group

Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



6. Business Recommendations

- **Boost Subscription Adoption**

  Promote exclusive perks such as early access, special discounts, and faster delivery to convert regular customers into subscribers.

- **Strengthen Customer Loyalty Programs**

  Introduce reward points, cashback, or tier-based benefits to encourage repeat purchases and transition customers into the “loyal” segment.

- **Optimize Discount Strategy**

  Evaluate the effectiveness of discounts and promo codes to ensure they drive sales without significantly impacting profit margins.

- **Enhance Product Positioning**

  Prioritize top-rated and high-performing products in marketing campaigns to increase visibility and conversions.

- **Implement Targeted Marketing**

  Focus campaigns on high-revenue customer segments, such as specific age groups and frequent express-shipping users, for better ROI.